

class05

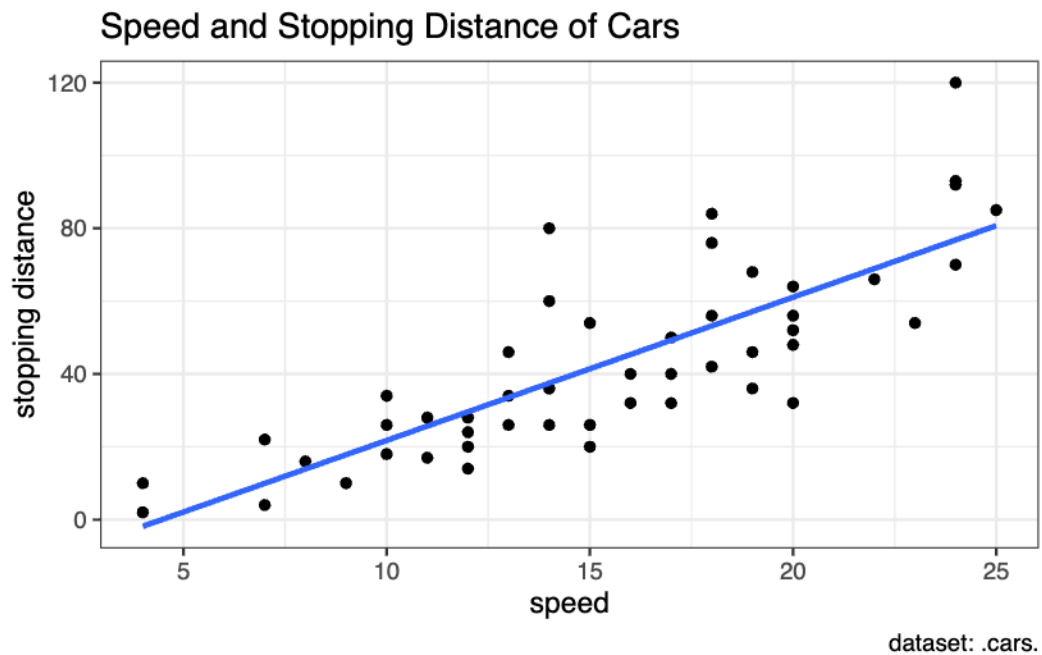
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Creating Scatter Plots

```
library(ggplot2)

ggplot(cars) + aes(x=speed, y=dist) + geom_point() +
  geom_smooth(method="lm", se=FALSE) +
  labs(title="Speed and Stopping Distance of Cars", x="speed",
       y="stopping distance", caption="dataset: cars ") + theme_bw()
```

`geom\_smooth()` using formula = 'y ~ x'



Q: Which geometric layer should be used to create scatter plots in ggplot2?

Answer: `geom_point()`

Q: In your own RStudio can you add a trend line layer to help show the relationship between the plot variables with the `geom_smooth()` function?

Answer: `geom_smooth()`

Q: Argue with `geom_smooth()` to add a straight line from a linear model without the shaded standard error region?

Answer: `geom_smooth(method="lm", se=FALSE)`

Q: Can you finish this plot by adding various label annotations with the `labs()` function and changing the plot look to a more conservative “black & white” theme by adding the `theme_bw` function?

A: `labs(title= “Speed and Stopping Distances of Cars”, x=“speed”, y=“stopping distance”, caption=“dataset: cars ”) + theme_bw()`

Adding more plot aesthetics through `aes()`

```
url <-  
  "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"  
genes <- read.delim(url)  
head(genes)
```

	Gene	Condition1	Condition2	State
1	A4GNT	-3.6808610	-3.4401355	unchanging
2	AAAS	4.5479580	4.3864126	unchanging
3	AASDH	3.7190695	3.4787276	unchanging
4	AATF	5.0784720	5.0151916	unchanging
5	AATK	0.4711421	0.5598642	unchanging
6	AB015752.4	-3.6808610	-3.5921390	unchanging

Q: Use the `nrow()` function to find out how many genes are in this dataset. What is your answer?

A: `nrow(genes) -> 5196`

Q: Use the `colnames()` function and the `ncol()` function of the genes data frame to find out what the column names are and how many columns there are. How many columns did you find?

A: `colnames(genes) -> "Gene", "Condition1", "Condition2", "State"`

A: `ncol(genes) -> 4`

Use the `table()` function on the `State` column of this data.frame to find out how many up regulated genes there are. What is your answer?

A: `table(genes$State) -> 127 up`

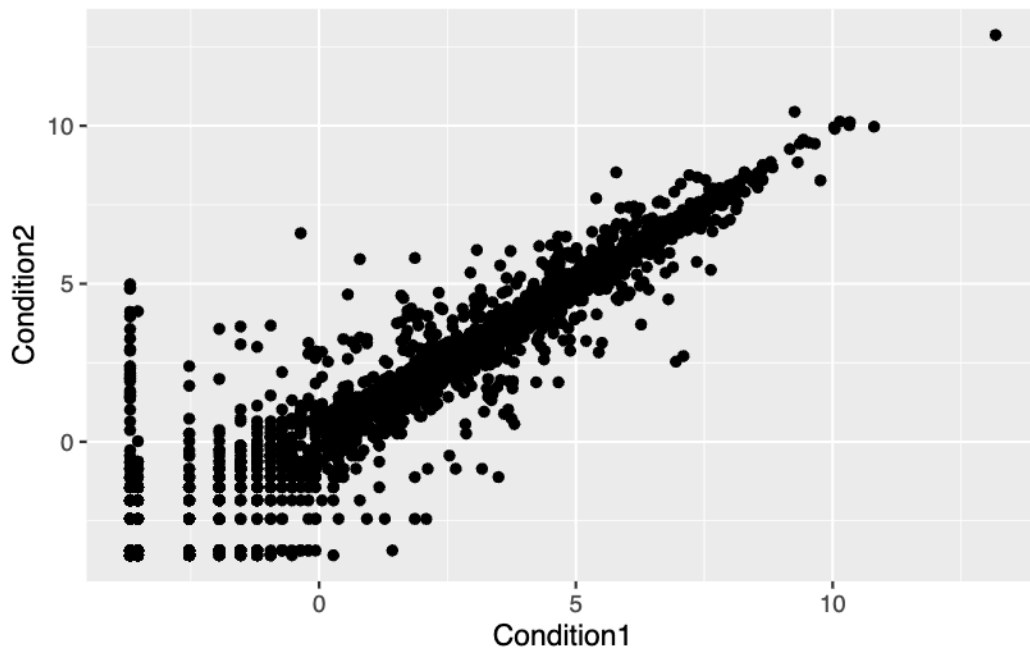
Using your values above and 2 significant figures. What fraction of total genes in up-regulated in this dataset?

A: `round(table(genes$State)/nrow(genes) * 100, 2) -> 2.44 up`

Q: Complete the code below to produce the following plot

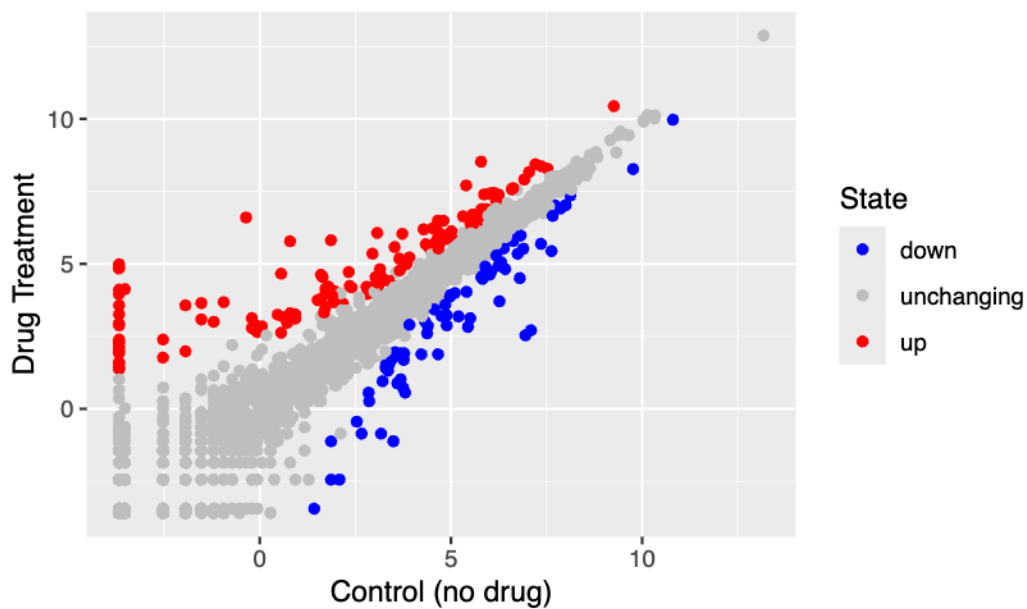
Answer:

```
ggplot(genes) + aes(x=Condition1, y=Condition2) + geom_point()
```



```
p <- ggplot(genes) + aes(x=Condition1, y=Condition2, col=State) + geom_point() +  
  labs(title="Gene Expression Changes Upon Drug Treatment",  
        x="Control (no drug)", y="Drug Treatment")  
p + scale_color_manual(values=c("blue", "gray", "red"))
```

Gene Expression Changes Upon Drug Treatment



Going Further

Q: Complete the code below to produce a first basic scatter plot of this gapminder_2007 dataset:

Answer:

```
library(gapminder)
library(dplyr)
```

Attaching package: 'dplyr'

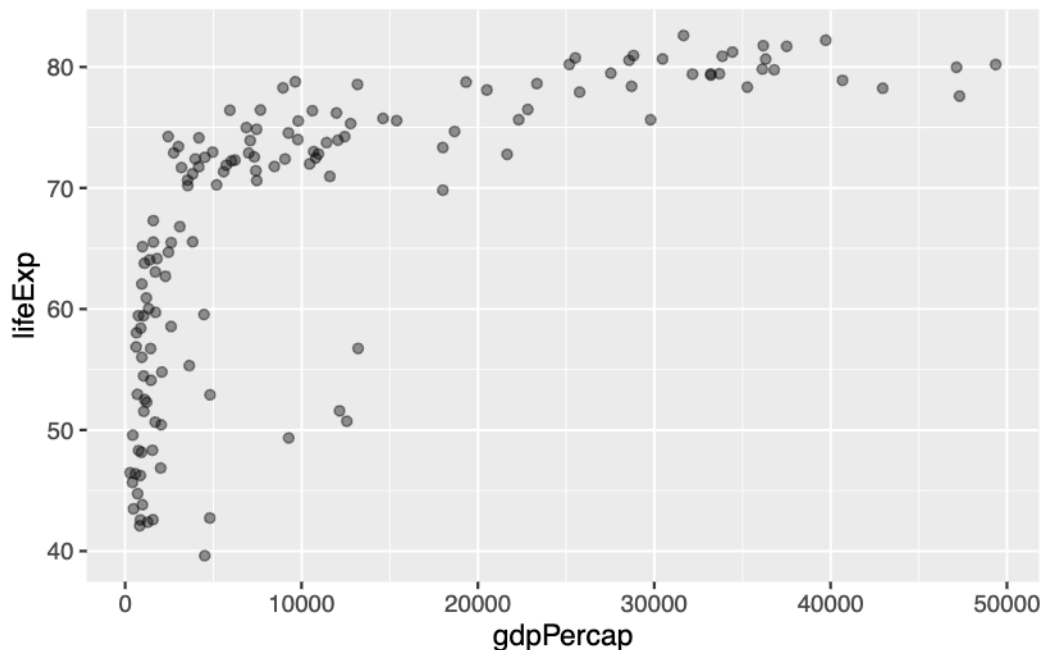
The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
gapminder2007 <- gapminder %>% filter(year==2007)
ggplot(gapminder2007) + aes(x=gdpPercap, y=lifeExp) + geom_point(alpha=0.4)
```



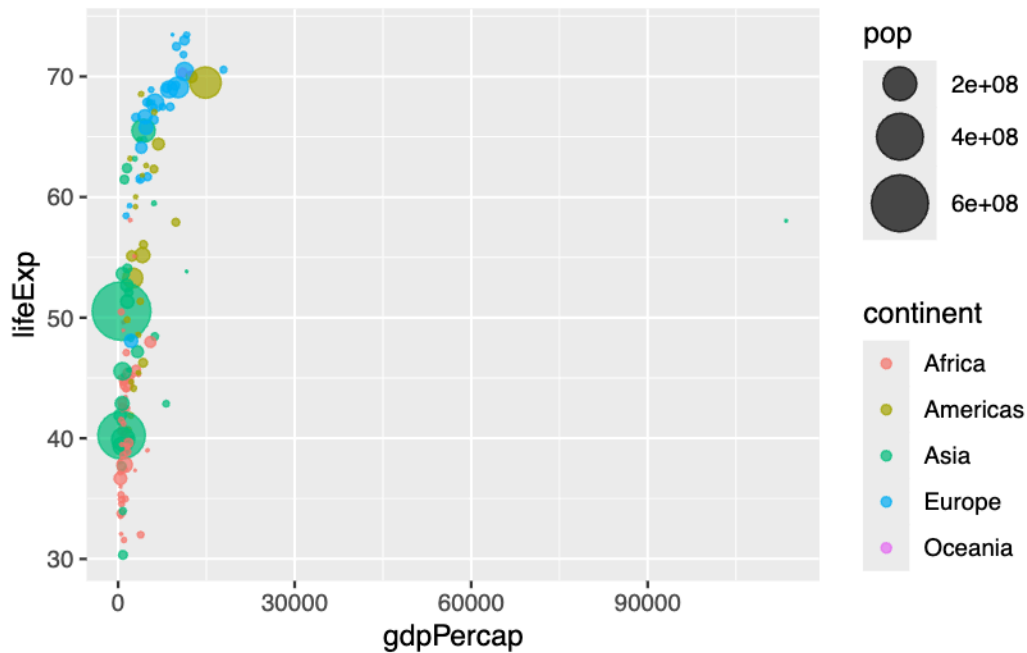
Adding more variables to aes()

Q: Can you adapt the code you have learned to reproduce our gapminder scatter plot for the year 1957? What do you notice about this plot is it easy to compare with the one for 2007?

A: Not very easy to compare because the graphs are scaled differently.

A:

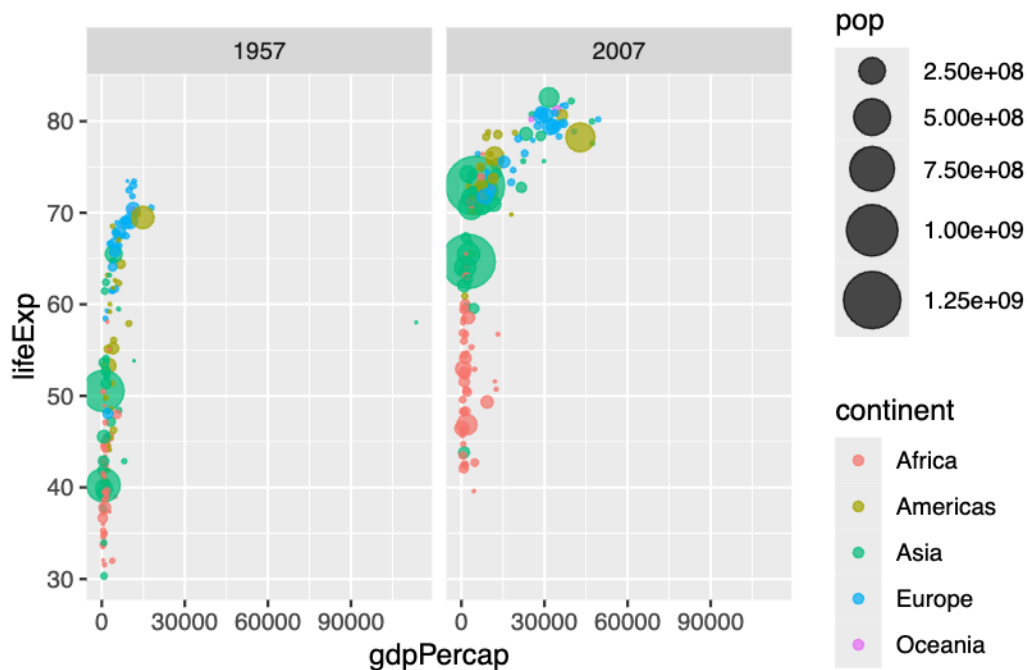
```
library(gapminder)
library(dplyr)
gapminder1957 <- gapminder %>% filter(year==1957)
ggplot(gapminder1957) + aes(x=gdpPercap, y=lifeExp, color=continent, size=pop) +
  geom_point(alpha=0.7) + scale_size_area(max_size=10)
```



Q: Do the same steps above but include 1957 and 2007 in your input dataset for ggplot(). You should now include the layer facet_wrap(~year) to produce the following plot:

```
gapminder1957_2007 <- gapminder %>% filter(year==1957 | year==2007)

ggplot(gapminder1957_2007) + aes(x=gdpPercap, y=lifeExp, color=continent,
  size=pop) + geom_point(alpha=0.7) + scale_size_area(max_size=10) +
  facet_wrap(~year)
```



Bar Charts

```
gapminder_top5 <-
  gapminder %>% filter(year==2007) %>% arrange(desc(pop)) %>% top_n(5, pop)

gapminder_top5
```

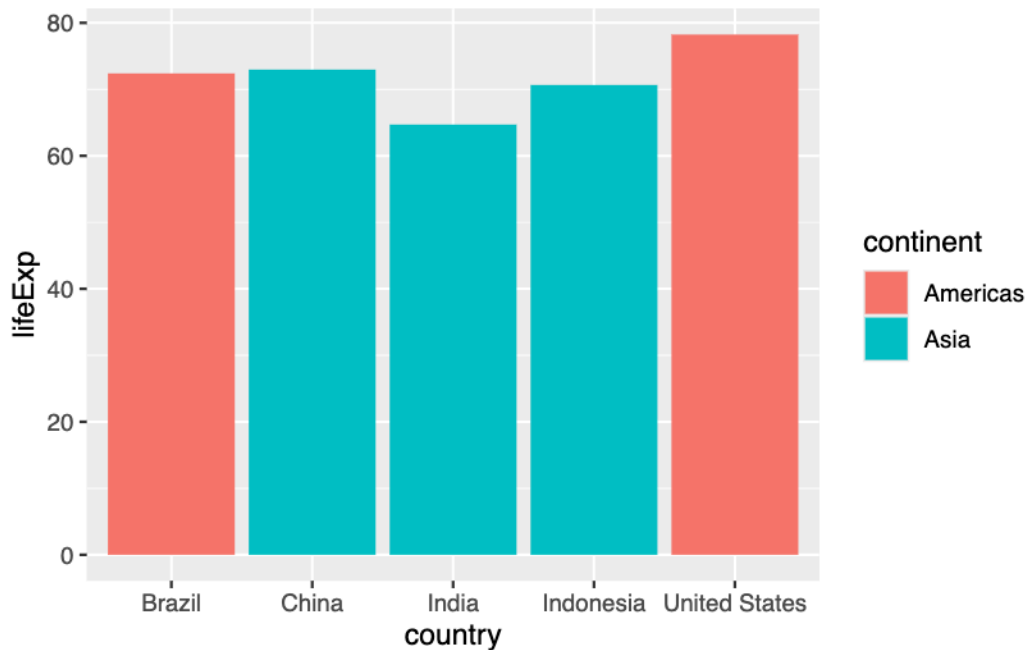
A tibble: 5 x 6

	country	continent	year	lifeExp	pop	gdpPercap
	<fct>	<fct>	<int>	<dbl>	<int>	<dbl>
1	China	Asia	2007	73.0	1318683096	4959.
2	India	Asia	2007	64.7	1110396331	2452.
3	United States	Americas	2007	78.2	301139947	42952.
4	Indonesia	Asia	2007	70.6	223547000	3541.
5	Brazil	Americas	2007	72.4	190010647	9066.

Creating a simple bar chart

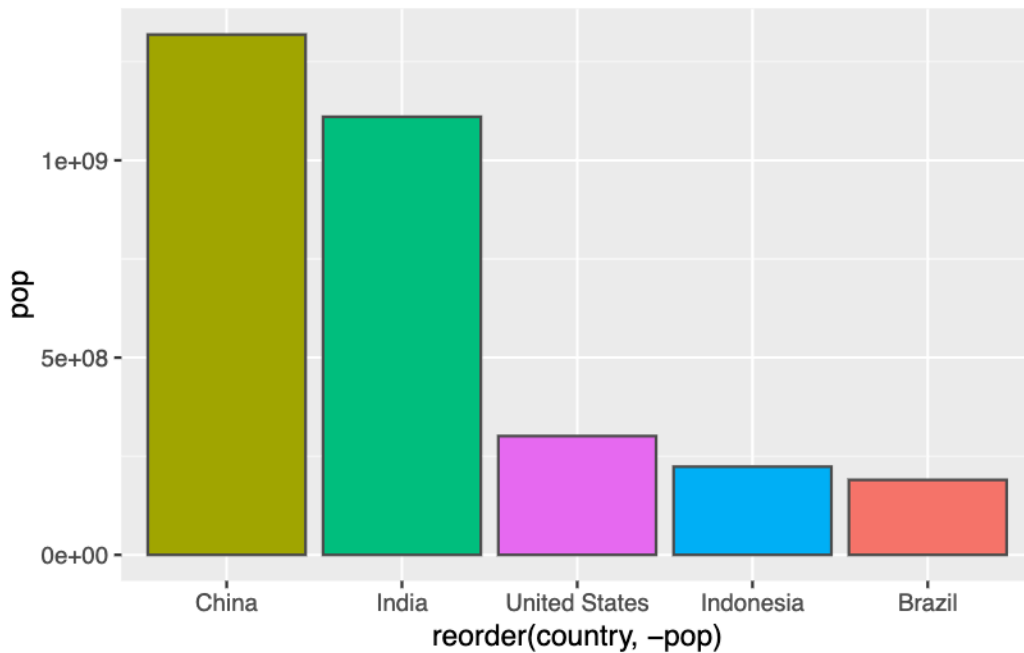
Q: Create a bar chart showing the life expectancy of the five biggest countries by population in 2007

```
ggplot(gapminder_top5) + geom_col(aes(x=country, y=lifeExp, fill=continent))
```



Q: Plot population size by country. Create a bar chart showing the population (in millions) of the five biggest countries by population in 2007

```
ggplot(gapminder_top5) + aes(x=reorder(country, -pop), y=pop, fill=country) +  
  geom_col(col="gray30") + guides(fill="none")
```

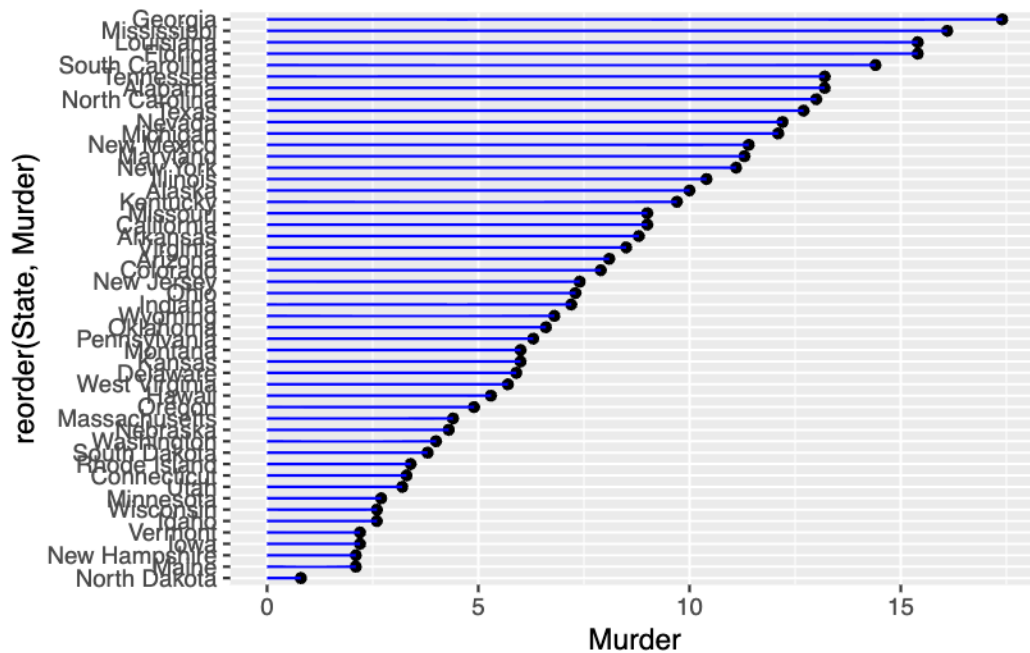


Flipping bar charts

```
head(USArrests)
```

	Murder	Assault	UrbanPop	Rape
Alabama	13.2	236	58	21.2
Alaska	10.0	263	48	44.5
Arizona	8.1	294	80	31.0
Arkansas	8.8	190	50	19.5
California	9.0	276	91	40.6
Colorado	7.9	204	78	38.7

```
USArrests$State <- rownames(USArrests)
ggplot(USArrests) + aes(x=reorder(State,Murder), y=Murder) + geom_point() +
  geom_segment(aes(x=State, xend=State, y=0, yend=Murder), color="blue") +
  coord_flip()
```



Extentions: Animation

The following code was used to render the animation:

```
ggplot(gapminder, aes(gdpPercap, lifeExp, size = pop, colour = country)) + geom_point(alpha = 0.7, show.legend = FALSE) + scale_colour_manual(values = country_colors) + scale_size(range = c(2, 12)) + scale_x_log10() + facet_wrap(~continent) + labs(title = 'Year: {frame_time}', x = 'GDP per capita', y = 'life expectancy') + transition_time(year) + shadow_wake(wake_length = 0.1, alpha = FALSE)
```

Combining plots

```
library(patchwork)

p1 <- ggplot(mtcars) + geom_point(aes(mpg, disp))
p2 <- ggplot(mtcars) + geom_boxplot(aes(gear, disp, group = gear))
p3 <- ggplot(mtcars) + geom_smooth(aes(disp, qsec))
p4 <- ggplot(mtcars) + geom_bar(aes(carb))
```

```
(p1 | p2 | p3) /  
  p4
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

